Aankit Das

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On F1 OPT STEM, Eligible to work in the US without sponsorship | TX

Work History

AI Engineer at Bantrly, Dallas, TX — *Mar 2026 to Present*

- Built the speech and language systems behind a K-12 learning app by turning student audio into text and scoring it against teaching rubrics
- Built a modular speech-to-text routing engine in Python and FastAPI that separated transcription from LLM scoring across 8 K-12 assessment modules, cutting audio API costs and improving rubric accuracy
- Built a two-layer voice activity detection pipeline combining silence-boundary audio chunking, timestamp deduplication, and a sliding guardrail text window, eliminating false positives during real-time streaming
- Built a provider-agnostic adapter registry across Parakeet TDT, Whisper, AssemblyAI, and Gemini with quality-based fallback and config-driven session health checks, letting engines swap with zero code changes

Machine Learning Researcher at University of Texas at Dallas, Richardson, TX — *Jun 2025 to Mar 2026*

- Built and optimized neural network pipelines to extract clean signal from noisy space telescope data for an NSF-funded cosmology research project
- Processed and managed 1,600 high-resolution CMB simulated maps totaling 24GB across multiple splits for model training
- Automated hyperparameter sweeps using Hydra and shell scripting to benchmark model performance across ells, detectors, and feature configurations at multiple resolutions

Data Scientist at SEI Investments Company, Kolkata, WB — *Oct 2021 to Aug 2023*

- Built automation tools and data workflows for a financial services team, turning manual document and reconciliation processes into faster digital systems
- Developed a Python-based prototype for automating document-filling using NLP libraries and OCR technology, reducing manual processing by 50 percent and improving data availability
- Implemented error-handling mechanisms and documented the algorithm design, implementation steps, and performance metrics, improving reliability and knowledge sharing across the team

Projects

Bantrly Lesson Generator, K-12 ELA Content Generation System

- Built an AI system that generates K-12 speaking lessons by combining a large language model with rule-based checks for safety and quality
- Built a hybrid LLM and rule-based lesson generation system for K-12 ELA speaking practice, combining Groq inference, a CCSS-aligned 80-skill taxonomy, few-shot RAG, and a 9-step validated pipeline to produce structured, grade-appropriate lessons from three inputs
- Designed multi-layer guardrails combining pre-generation input validation, post-generation content checks for vocabulary ceiling and cognitive load, and a cultural bias retry loop that fed flagged terms back into the prompt across up to 5 retries, ensuring no biased lesson was ever saved

• Engineered a defensive validation pipeline for non-deterministic LLM output, auto-correcting three distinct practice field structures across model variants with reflect-stage normalization and domain-specific field stripping to ensure clean JSON regardless of model behavior
• Deployed a 6-tab Gradio UI on Hugging Face Spaces combining a student-facing step-by-step lesson experience, a coverage heatmap, a skill taxonomy browser, and a guardrail inspector, giving every view live updates after each generation

Document Intelligence Platform, Production RAG System
• Built a retrieval-augmented generation system that lets users ask questions in plain language and get answers grounded in their own documents
• Engineered a production-grade retrieval-augmented generation system combining local embedding models Ollama and Sentence-Transformers, Chroma DB vector storage, and the Groq llama-3.1-8b-instant LLM API, delivering intelligent document question-answering
• Designed a modular Python architecture of 5 independent modules, chunker, embeddings, vector store, llm, and pipeline, improving testability and maintainability aligned with software engineering best practices
• Implemented FastAPI REST endpoints with async request handling, Pydantic type safety, health checks, and structured logging for production observability
• Deployed a containerized application using Docker multi-stage builds and automated CI/CD via GitHub Actions, hosting it on Hugging Face Spaces with serverless scaling

Education & Certificates
• MS, Computer Engineering, University of Texas at Dallas (GPA - 3.8/4.0) May 2025
• BTech, Electronics and Communication Engineering, University of Calcutta (GPA - 3.7/4.0) Sept 2020

Publications
• **Das, A.** A hybrid meta-heuristic feature selection method for identification of Indian spoken languages from audio signals. IEEE Access.
• **Das, A.** EmergeNet: A novel deep-learning based ensemble segmentation model for emergence timing detection of coleoptile. Frontiers in Plant Science.
• **Das, A.** Hybrid feature selection method based on harmony search and naked mole-rat algorithms for spoken language identification from audio signals. IEEE Access.
• **Das, A.** Flowerphenonet: Automated flower detection from multi-view image sequences using deep neural networks for temporal plant phenotyping analysis. MDPI.